

SIH-2026 Gap Analysis Report

Project Mansakha - SIH 2026

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Version 1.0

SIH-2026 Problem Statement:

Meticulous Gap Analysis

Following a strict, word-by-word evaluation of the provided Problem Statement against the current codebase, here is the exhaustive list of missing features, non-functioning stubs, and critical vulnerabilities.



1. The Demographic & Accessibility Gap

"...Beneficiaries receiving relief under the Scheduled Castes and Scheduled Tribes (Prevention of Atrocities) Act..."

The Gap: We have built a sophisticated React Native Smartphone App, but many rural or marginalized SC/ST victims rely on feature phones.

"...through chatbot, IVRS calls, SMS, mobile applications, web portal, or helpline follow-up mechanisms."

The Gap: The codebase over-indexes on the Smartphone App. The low-tech, high-accessibility channels are completely non-functioning: **IVRS (Interactive Voice Response):** Stubbed in `dispatchWorker.js`. It explicitly throws an error: `IVRS provider configured but call placement is not yet implemented`. **SMS Prompts:** Stubbed. Attempting to send an SMS check-in throws an error. **Helpline Follow-Up:** There is no UI for call center agents to log follow-up calls or integrate with the 14566 helpline.



2. The Privacy & Security Vulnerability

"...Ensure explainable AI, privacy protection, data security, and compliance..."

The Gap (Critical Privacy Flaw): When a victim types a journal entry or chat message (e.g., "My neighbor Ramesh threatened me today at my house in Pune"), the backend (`services/ai.js`) sends this **raw, unredacted text** directly to the Google Gemini API over the internet. There is **zero PII (Personally Identifiable Information) scrubbing** happening before transmission. Sending unredacted victim trauma data to a third-party LLM is a massive violation of data security and privacy compliance.

3. Dummy / Simulated Features

"...registered through NHAA (14566), the Integrated Portal..."

The Gap (Dummy UI): The "Fetch Case Details" feature used by Data Operators to look up cases from the National Helpline (NHAA) is **a complete dummy feature**. The React UI (`FetchCase.jsx`) and the backend route (`/fetch-case`) explicitly state that this is Simulated Data. When a user types in a Docket Number, the backend runs a mathematical hash on the string to generate a fake Case Type and Stage. There are no actual webhooks or API integrations built to pull live data from the government portals.

4. Missing AI Capabilities

"...Analyse voice, text, behavioural responses, and engagement patterns using NLP, Sentiment Analysis, and Emotion AI."

The Gap (Voice): Voice Stress Analytics is completely missing. In `services/scoring.js`, the code explicitly sets `voiceStress: 0` and notes it is "deferred for Phase 2."

"...Ensure explainable AI..."

The Gap (Explainability): The AI currently generates a "reasoning" for why it gave a high distress score. It saves this to the database (`distress_scores.explanation`).

However, this explanation is **hidden from the Counsellors** in the React UI, rendering the AI unexplainable to the human workers who actually need to understand it.

5. Reactive vs. Predictive

"...Predict escalation of psychological distress before a crisis situation emerges."

The Gap: The current math in `scoring.js` is purely **reactive**. It calculates a score based on a current message, and if it crosses 80 (Critical), it fires an alert. It does not actively analyze a victim's historical trajectory to predict that they will hit 80 next week (which would be true "prediction before a crisis emerges").

6. Missing Case Tracking Integrations

*"...prolonged psychological distress after complaint registration due to threats, intimidation, repeated court appearances, delays in investigation and trial..."**

The Gap: While the database has a `case_stage` column (Investigation, Trial, etc.), the system does not actually track "court appearances" or "delays". There is no calendar system, no tracking of how many days a trial has been delayed, and the AI prompt is not contextually aware of upcoming court dates to preemptively check on the victim's stress.